Machine Learning Library Imports

A structured compilation of essential Python libraries for classification pipelines, evaluation, and class imbalance handling.

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, 
ConfusionMatrixDisplay
import matplotlib.pyplot as plt
import pandas as pd
import os
from imblearn.over_sampling import SMOTE
```

Component Breakdown:

- **Classifiers:** `LogisticRegression`, `RandomForestClassifier`, and `KNeighborsClassifier` for training distinct predictive models.
- **Preprocessing & Pipeline:** `StandardScaler` for feature normalization, and `Pipeline` to seamlessly chain preprocessing steps with estimators.
- **Model Selection & Evaluation:** `train_test_split` to split datasets, alongside `classification_report` and `ConfusionMatrixDisplay` for comprehensive performance metrics.
- **Imbalance Handling:** `SMOTE` (Synthetic Minority Over-sampling Technique) to balance target class distributions.